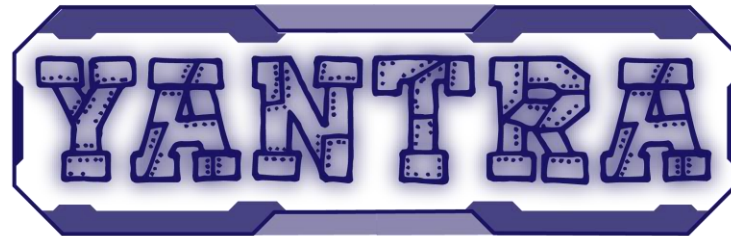


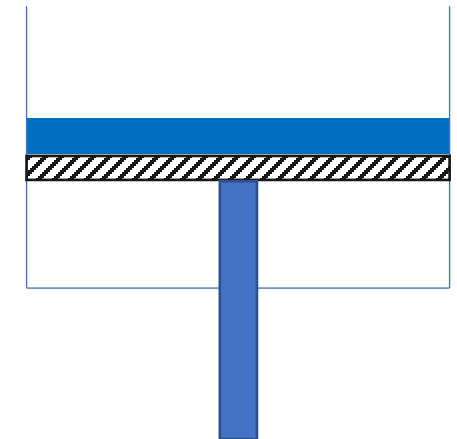
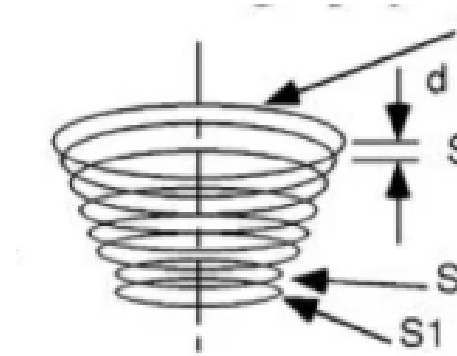
Solid ground curing



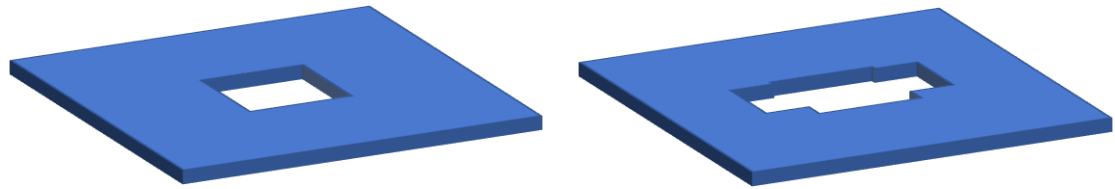
2. Solid ground curing

Accordingly, the process has **two** main cycles,
the **photomask generation cycle**
and a layer **fabrication cycle**.

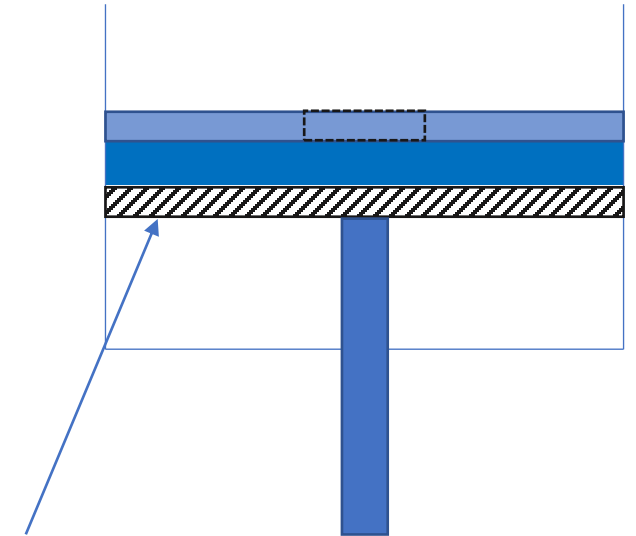
It takes about 2 min to complete all
operations to make a single layer.



Solid ground curing



photomask

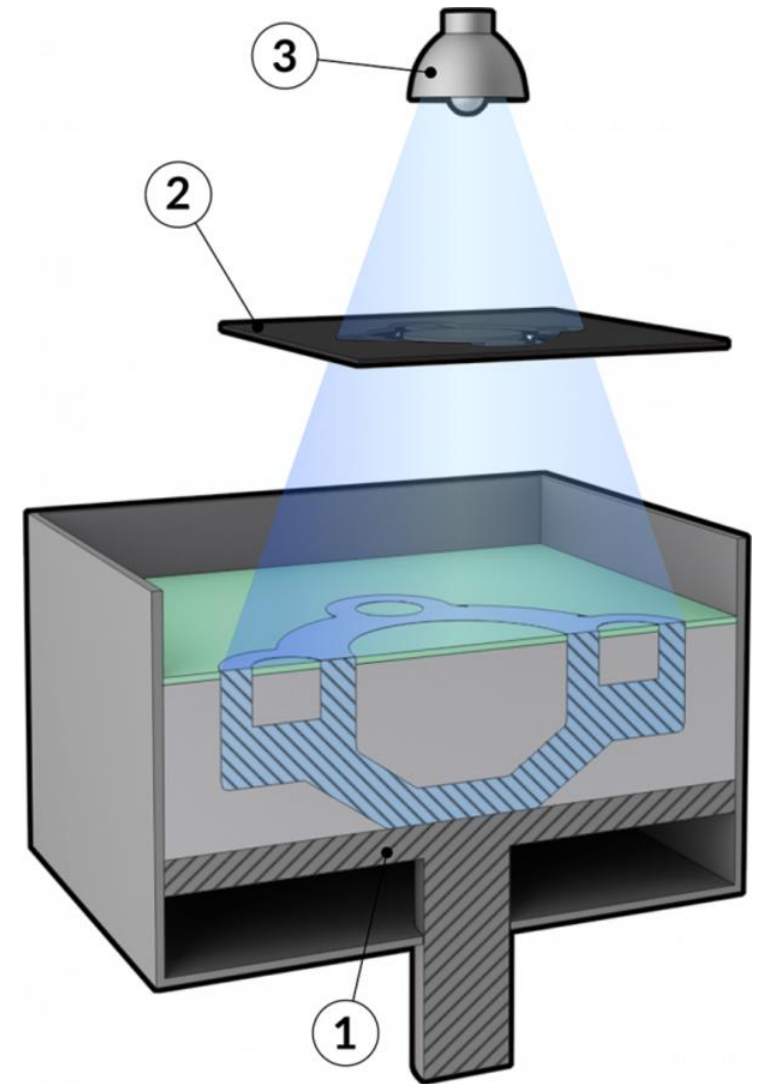


platform

PRINCIPLE

1. Parts are built, layer by layer, from a liquid photopolymer resin that solidifies when exposed to UV light.

The photo polymerization the irradiation source is a high power collimated UV lamp and the image of the layer is generated by masked illumination instead of optical scanning of a laser beam.

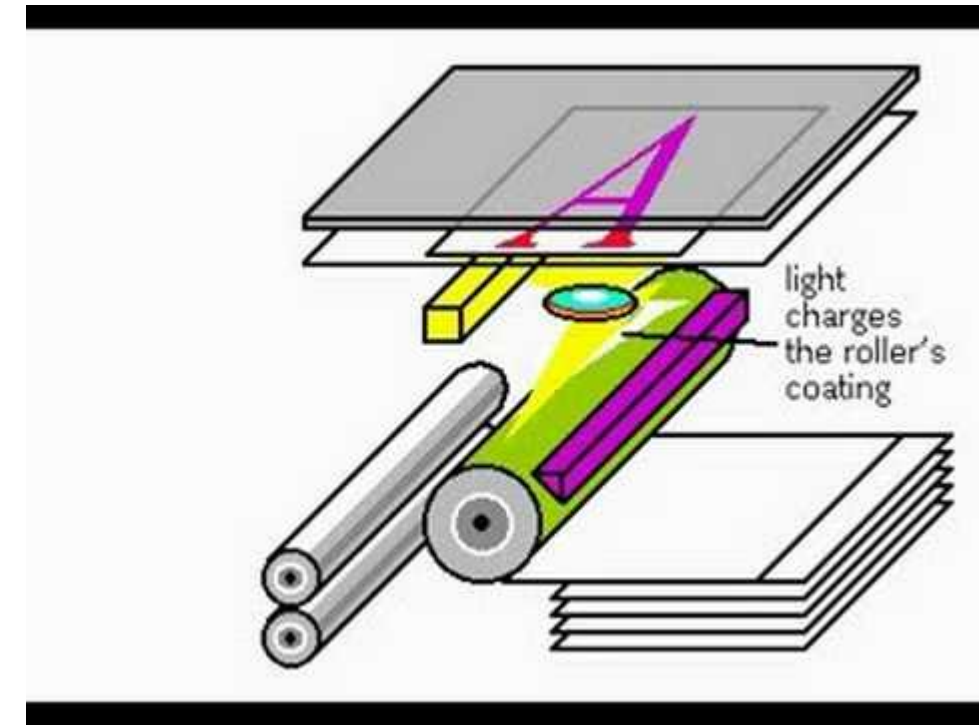


The **mask is created** from the CAD data input and “printed” on a transparent substrate (the mask plate) by an nonimpact ionographic printing process, a process similar to the

Xerography process used in photocopiers and laser printers.

This is used to mask the uniform illumination of the UV lamp.

After exposure, the electrostatic toner is removed from the substrate for reuse and the pattern for the next layer is similarly “printed” on the substrate.



photomask generation

Xerography

Meaning

Substrate

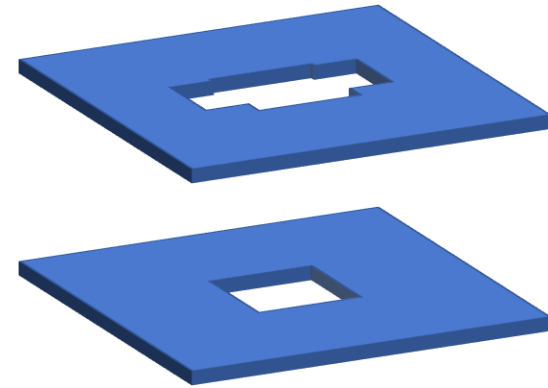
an underlying
substance or layer

2. Multiple parts may be processed and built in parallel by grouping them into batches (runs)

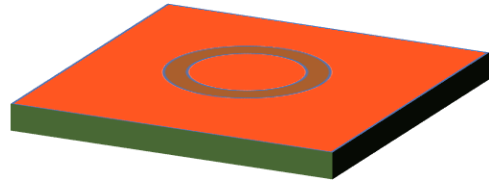
3. Each layer of a multiple layer run contains cross-sectional slices of one or many parts. Therefore, all slices in one layer are created simultaneously.

The next layer is then built immediately on the top of the created layer.

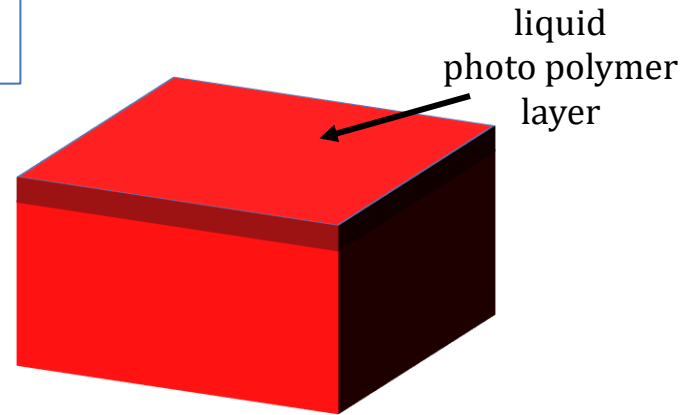
4. The process is **self-supporting** and does not require the addition of external support structures to emerging parts since continuous structural support for the parts is provided by the use of wax, acting as a solid support material.



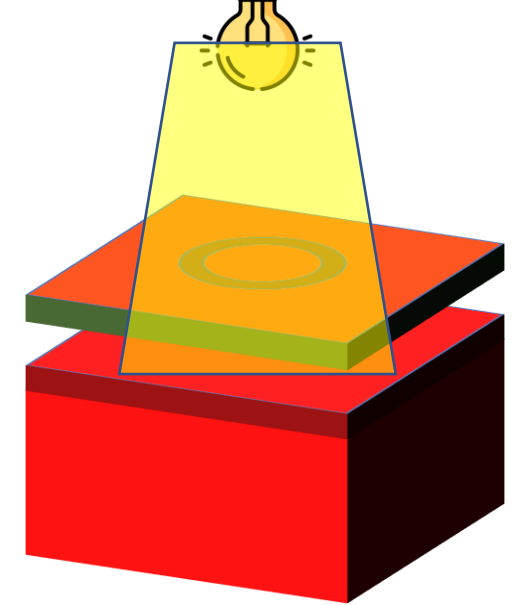
Process



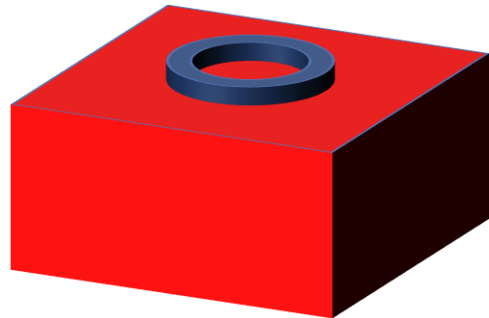
Mask Preparation



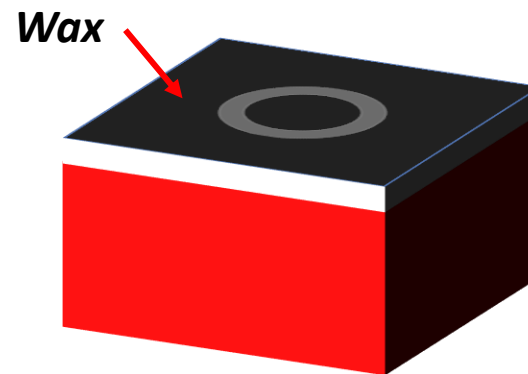
Applying liquid photo polymer layer



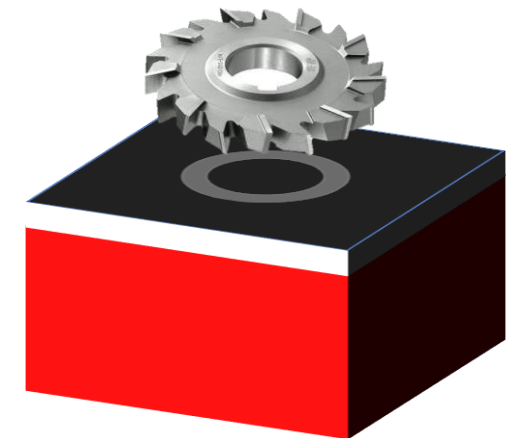
Mask Positioning and exposure of layer



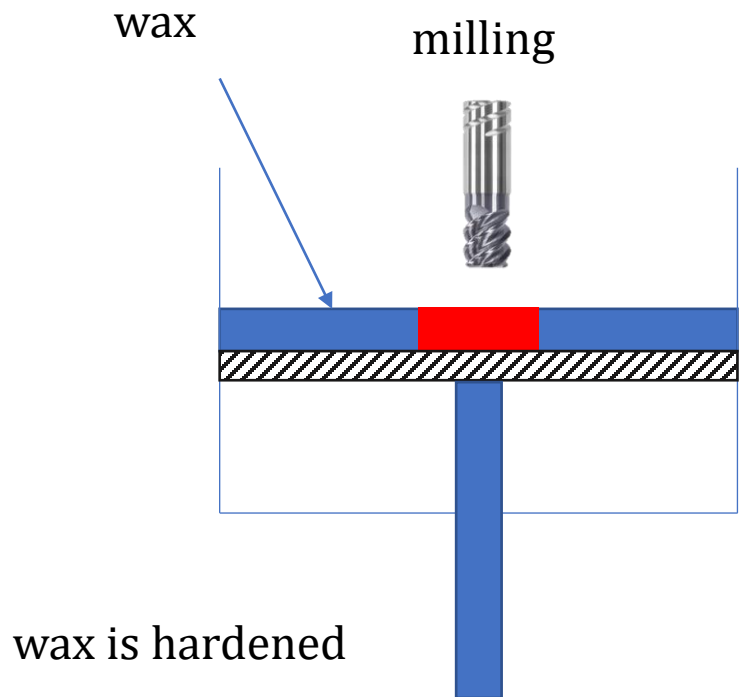
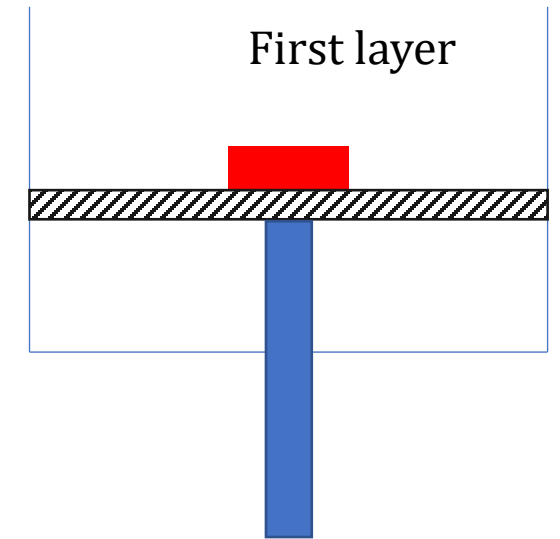
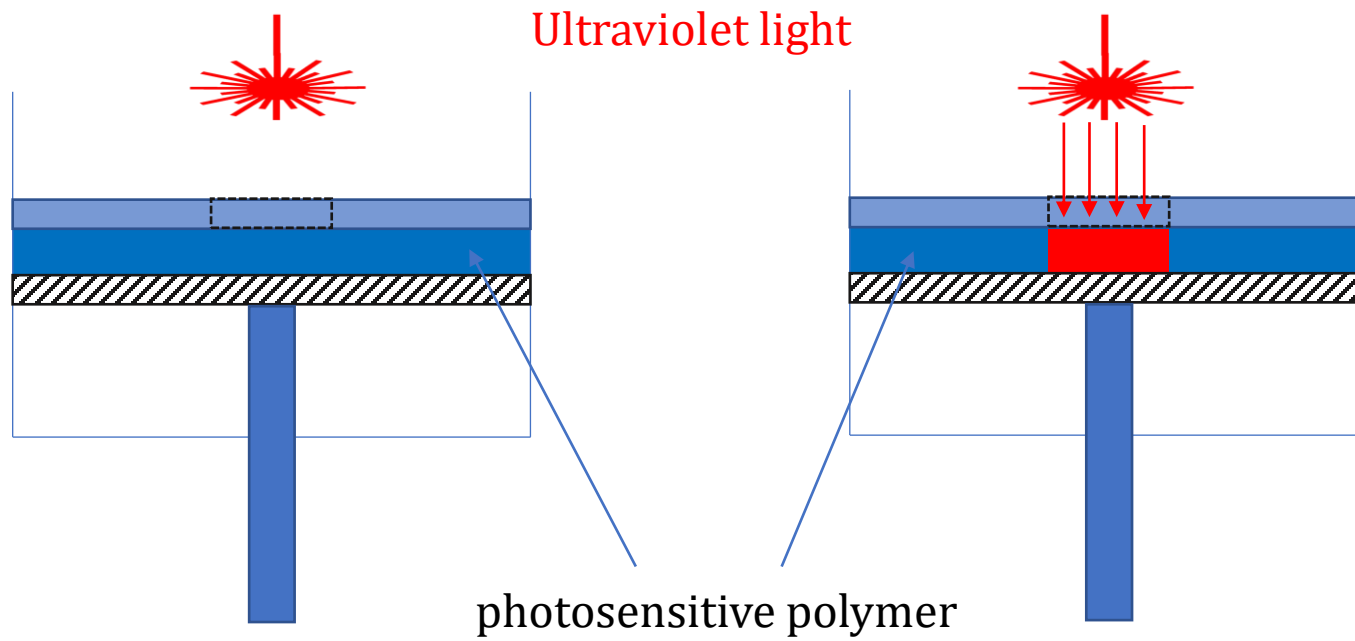
Uncured polymer is removed from the surface



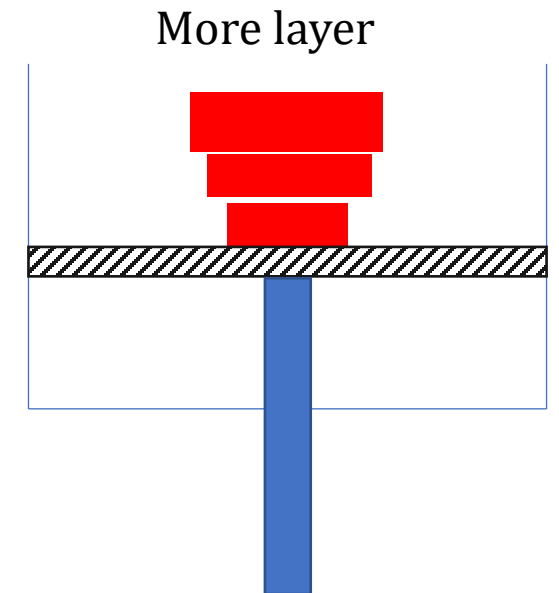
Wax filling



Milling for flatness and thickness



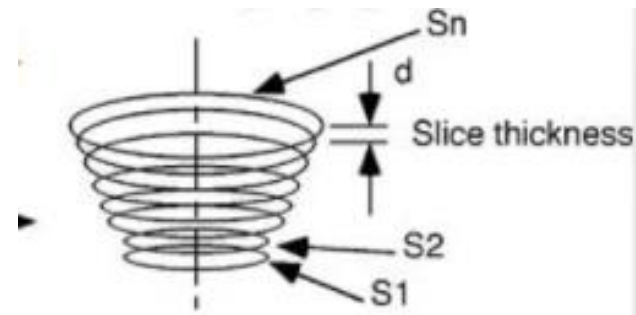
Procedure is repeated to form more layers



Alternate diagram

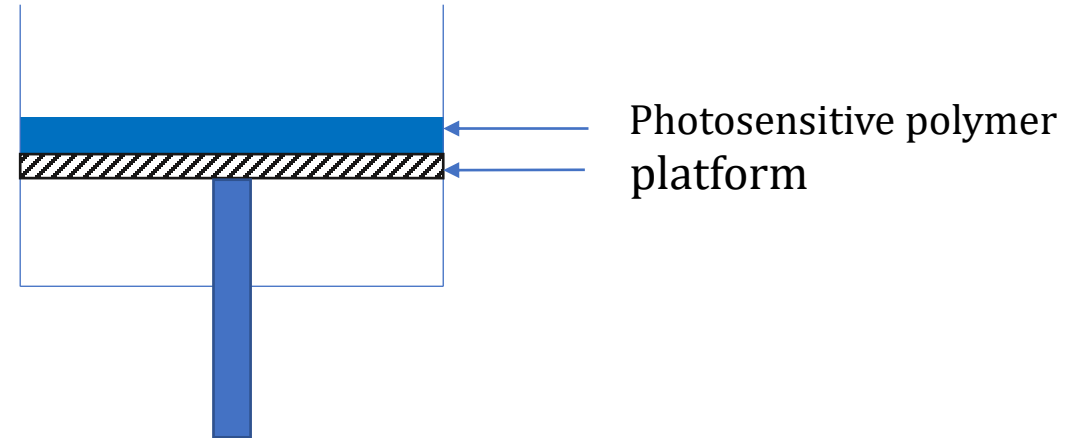
The following steps are followed during SGC

1. CAD files are **sliced**.

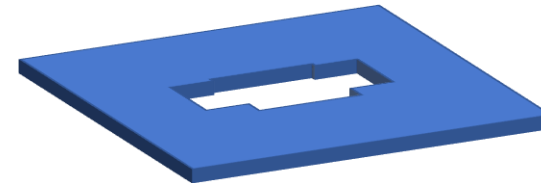


Slicing to N layers data

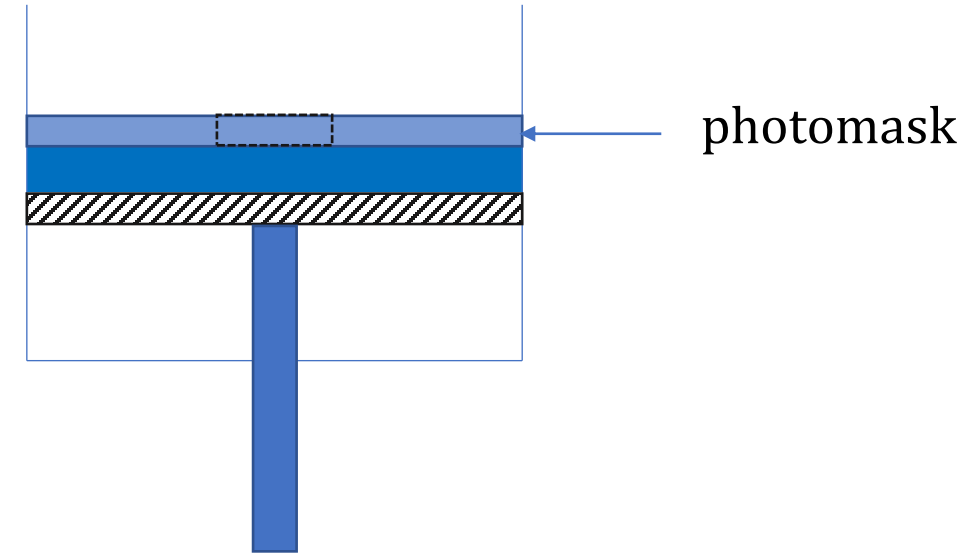
2. A layer of photosensitive polymer is first sprayed onto the platform.



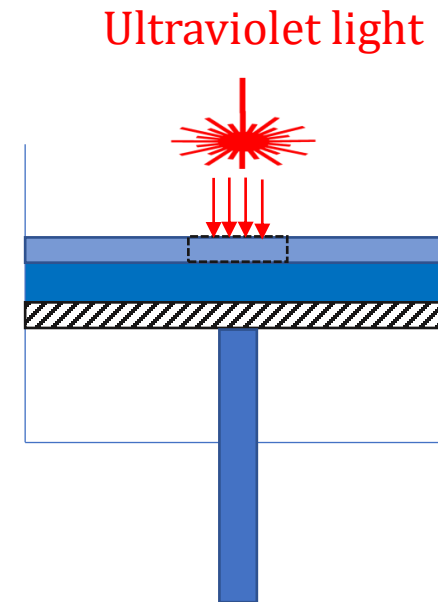
3. A photomask is prepared (for each layer) using **electrophotography** (xerography).



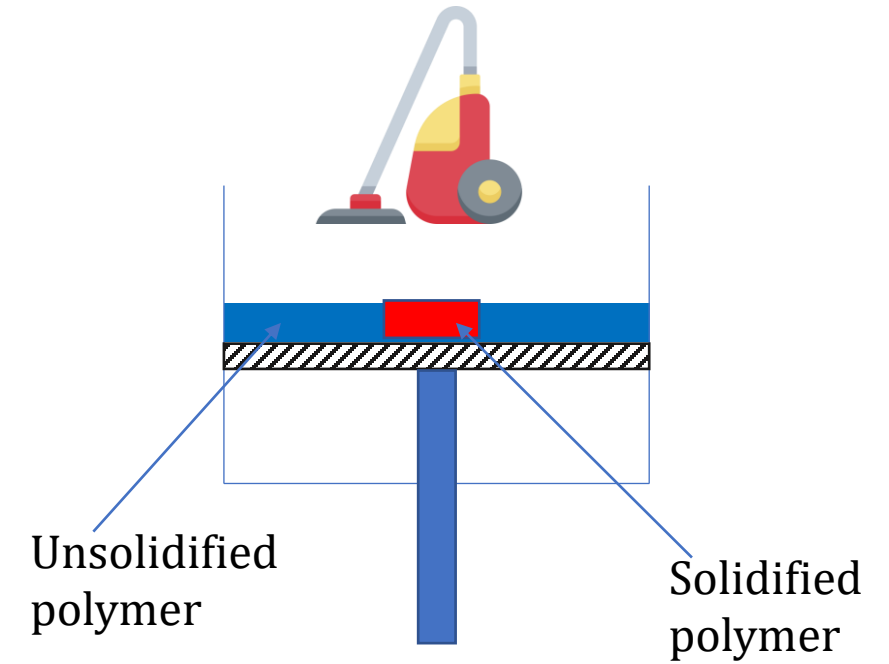
4. The **mask plate** is then placed on top of the platform and object under construction.



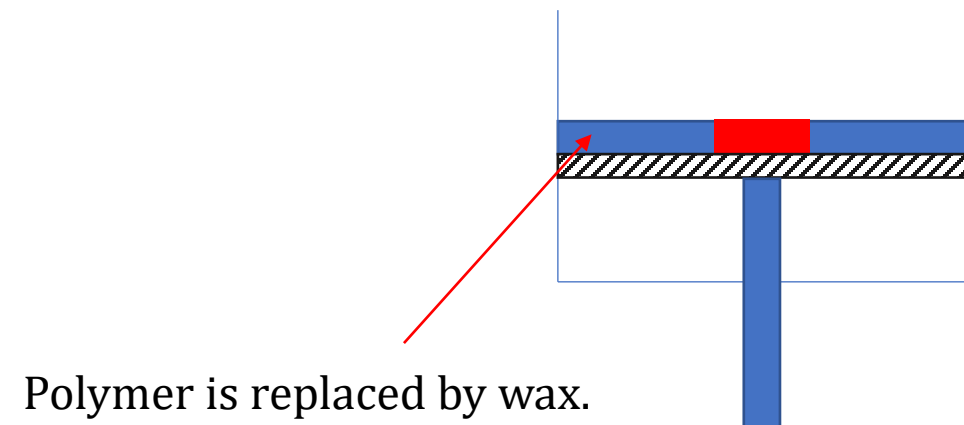
5. **Ultraviolet light** is exposed onto the layer, which solidifies completely.



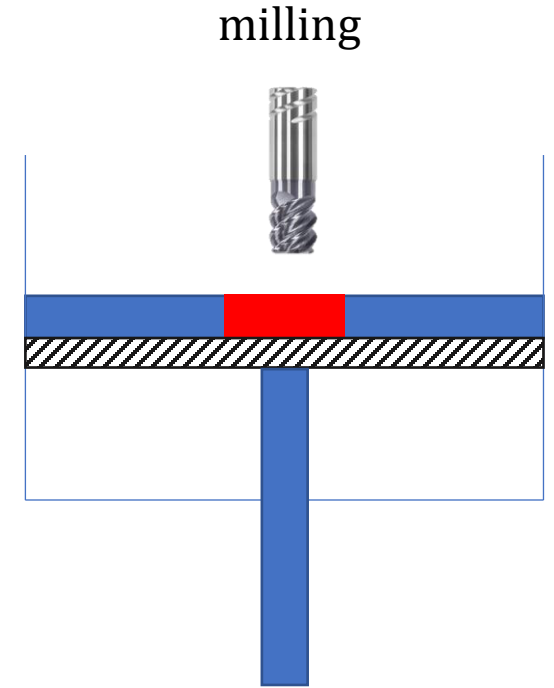
6. The residual polymer is **vacuumed** for recycling and **replaced by wax**.



7. The wax is hardened using the wax cooling plate



8. The **milling head** is used to machine the photopolymer to an accurate and uniform thickness in preparation for the next layer.

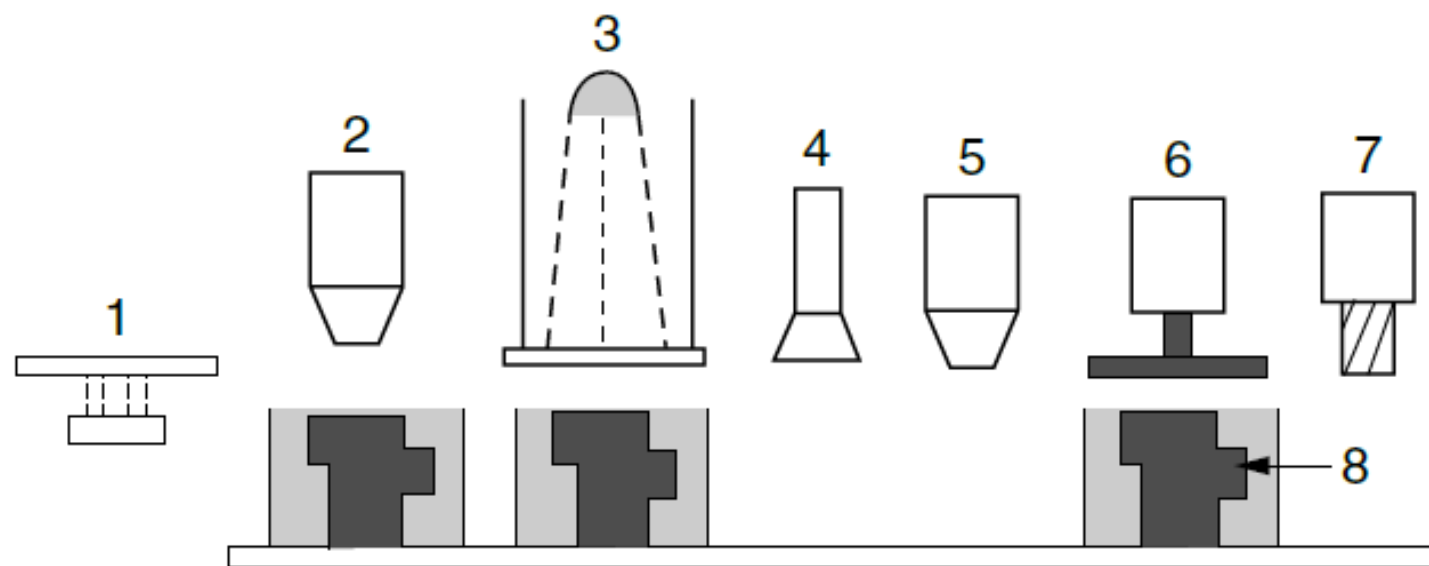


9. The mask plate is **erased and reused** after every layer.

10. The process is repeated for the next slice until the object is completely finished.



Final product



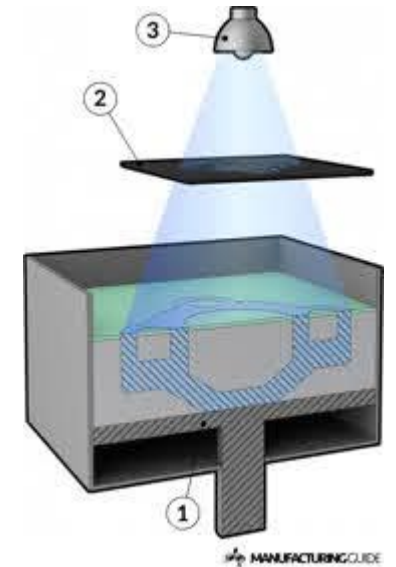
- 1 Mask preparation
- 2 Photopolymer resin application
- 3 UV light exposure
- 4 Aerodynamic wiper
- 5 Wax application
- 6 Wax cooling
- 7 Milling
- 8 Prototype

Figure 8.3 Solid ground curing steps.

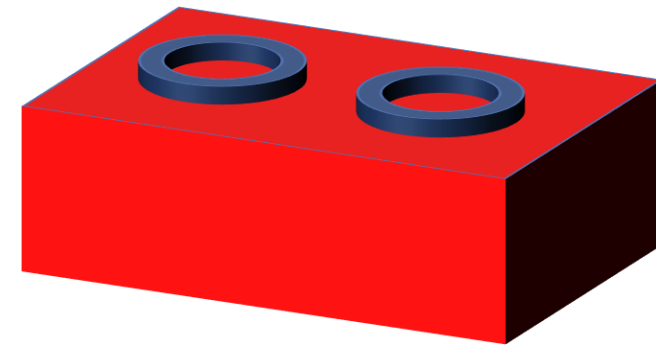
Advantages of SGS

Parallel processing: The process is based on instant, simultaneous curing of a whole cross-sectional layer area (rather than point-by-point curing). It has a high speed throughput that is about eight times faster than its competitors. Many parts can be produced at the same time.

- **Complicated** and large components are produced.
- **Self-supporting:** The solid **wax supports the part** in all dimensions. There is no need for additional supports.
- **High structural strength and stability:** This is due to the curing process that minimizes the development of internal stresses in the structure. As a result, they are much less brittle.



- **No post curing is needed**, which reduces internal stresses and warping.
- **Multiple parts** can be run at the **same time** due to the large workspace.
- **Accuracy** is good.
- Part complexity does not affect speed; however, the volume does.
- Weights may be inserted at any time to alter the center of gravity.
- **Layers can be milled** off if they are found to be in **error**.
- **Minimum shrinkage effect**: This is due to the full curing of every layer.



Multiple parts

Disadvantages of SGS

- The **cost** of equipment is high.
- The machine size is **large**.
- The process is rather complicated and requires high maintenance and **skilled supervision**.
- The range of materials is limited.
- The process is **noisy**.
- **Overexposure** of the polymer may increase the viscosity and **make it unusable**, thus greatly increasing the volume of expensive polymers used.
- The resins require that light sealed chambers and toxic material handling procedures be used.

APPLICATIONS

1. General applications: Conceptual design presentation, design proofing, engineering testing, integration and fitting, functional analysis, exhibitions and pre-production sales, market research, and inter-professional communication.
2. **Tooling and casting applications:** Investment casting, sand casting, and rapid, toolfree manufacturing of plastic parts.
3. **Mold and tooling:** Silicon rubber tooling, epoxy tooling, spray metal tooling, acrylic tooling, and plaster mold casting.
4. **Medical imaging:** Diagnostic, surgical, operation and reconstruction planning and custom prosthesis design.